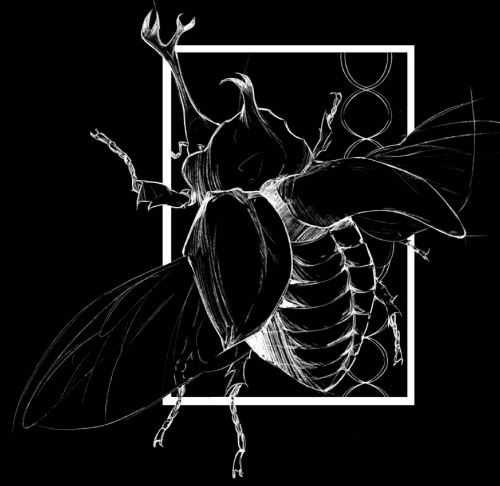




Abstract

Genome size and chromosome number are fundamental characteristics of organismal complexity that vary dramatically across the fungal kingdom. Understanding the evolutionary relationship between these genomic traits is essential for comprehending how fungal genomes are organized and how they evolve. Genome size, measured in megabases (Mb), reflects the total DNA content and is influenced by transposable element (TE) abundance, heterochromatin content, and polyploidy events. Chromosome number, by contrast, reflects how that DNA is organized into discrete units and can change through chromosomal rearrangements (fusions, fissions), polyploidy, and aneuploidy. While these traits might be expected to co-vary, the mechanisms driving their evolution may be partially independent. **This study investigates the relationship between genome size and chromosome number across 140 fungal species** representing major fungal lineages using phylogenetic comparative methods. By compiling comprehensive genomic data and employing rigorous statistical approaches that account for evolutionary non-independence among species, this study aims to determine whether genome size is a reliable predictor of chromosome number evolution in fungi.

Study Design & AI

This study used phylogenetic comparative methods to analyze genome size and chromosome number across 140 fungal species. Genomic data from published sources were placed on a comprehensive fungal phylogenetic tree and Generalized Least Squares (GLS) regression with Brownian motion covariance. I tested whether genome size predicts chromosome number while properly accounting for evolutionary relationships among species. This phylogenetic approach is essential because species are not independent data points and share evolutionary history through common ancestors. AI assisted with data organization, statistical analysis, and visualization. All methods are reproducible using open-source R software.

Discussion

Analysis reveals that genome size does not significantly predict chromosome number in fungi, suggesting these traits evolve through independent mechanisms rather than being functionally linked. Genome size is primarily driven by transposable element accumulation and heterochromatin expansion, while chromosome number changes result from chromosomal rearrangements (fusions and fissions) and polyploidy events processes that operate largely independently, explaining why larger-genome fungi don't necessarily have more chromosomes. The phylogenetic GLS model reversed the slope direction compared to standard regression, demonstrating that accounting for evolutionary history fundamentally changes understanding of the relationship. The strong phylogenetic signal for both traits ($\lambda \approx 0.9$) indicates that lineage history and evolutionary constraints matter more than simple trait correlations, and different fungal lineages appear to have evolved distinct solutions to genome organization without a consistent relationship to total genome size. This decoupling contrasts with some plant and animal systems and suggests fungi have evolved a uniquely independent system of genomic organization driven by different selective pressures on each trait.

Results

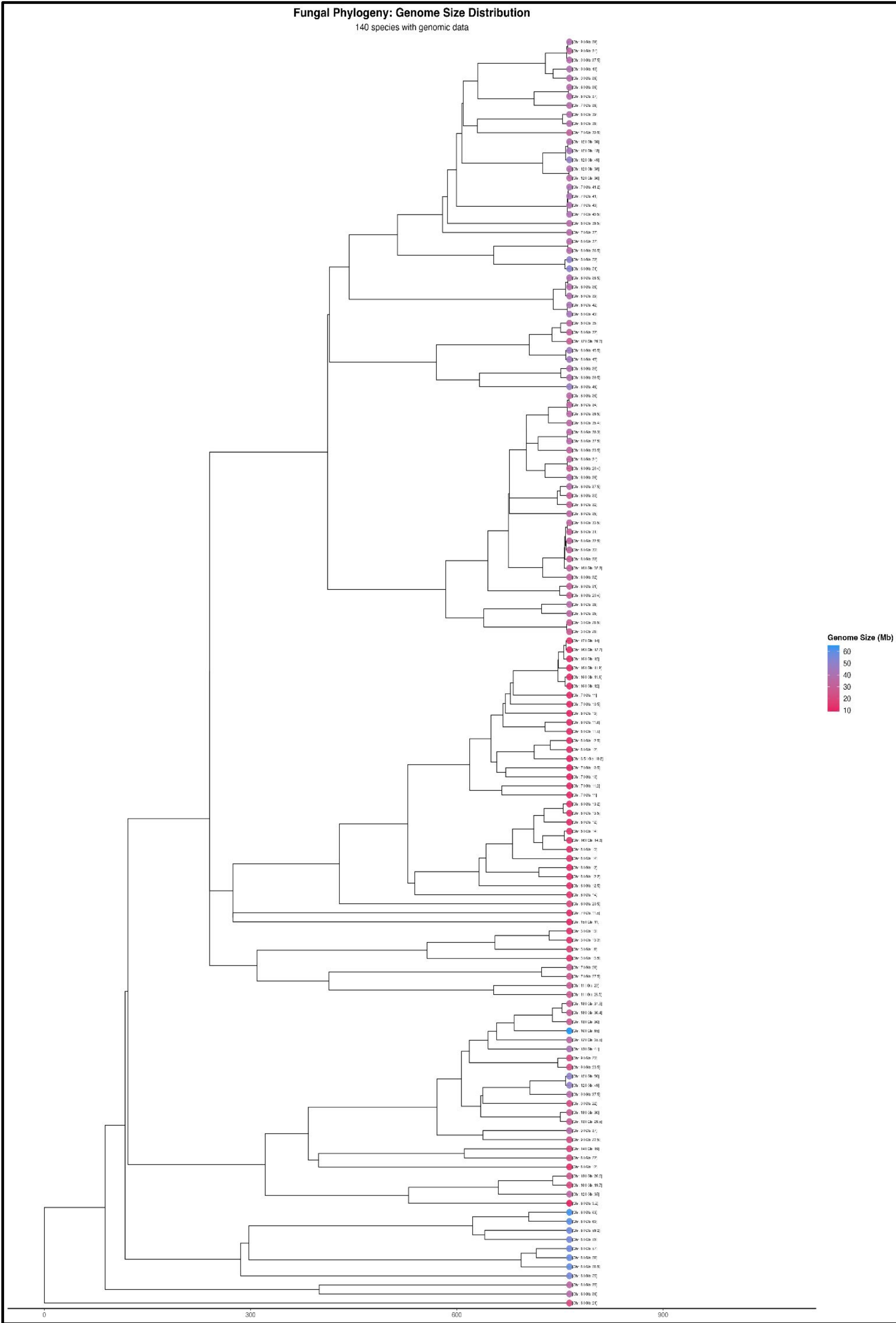


Figure 1: Phylogenetic Tree with Gradient-Colored Tips

A phylogenetic tree of 140 fungal species with branches representing evolutionary relationships. Species names appear at each tip, followed by trait values [Chr: # | Gb: #.#]. Tips are colored on a gradient from pink (low genome size) to blue (high genome size). Related species have similar colors, indicating strong phylogenetic signal. This visualization shows genome size variation follows evolutionary lineages rather than being random.

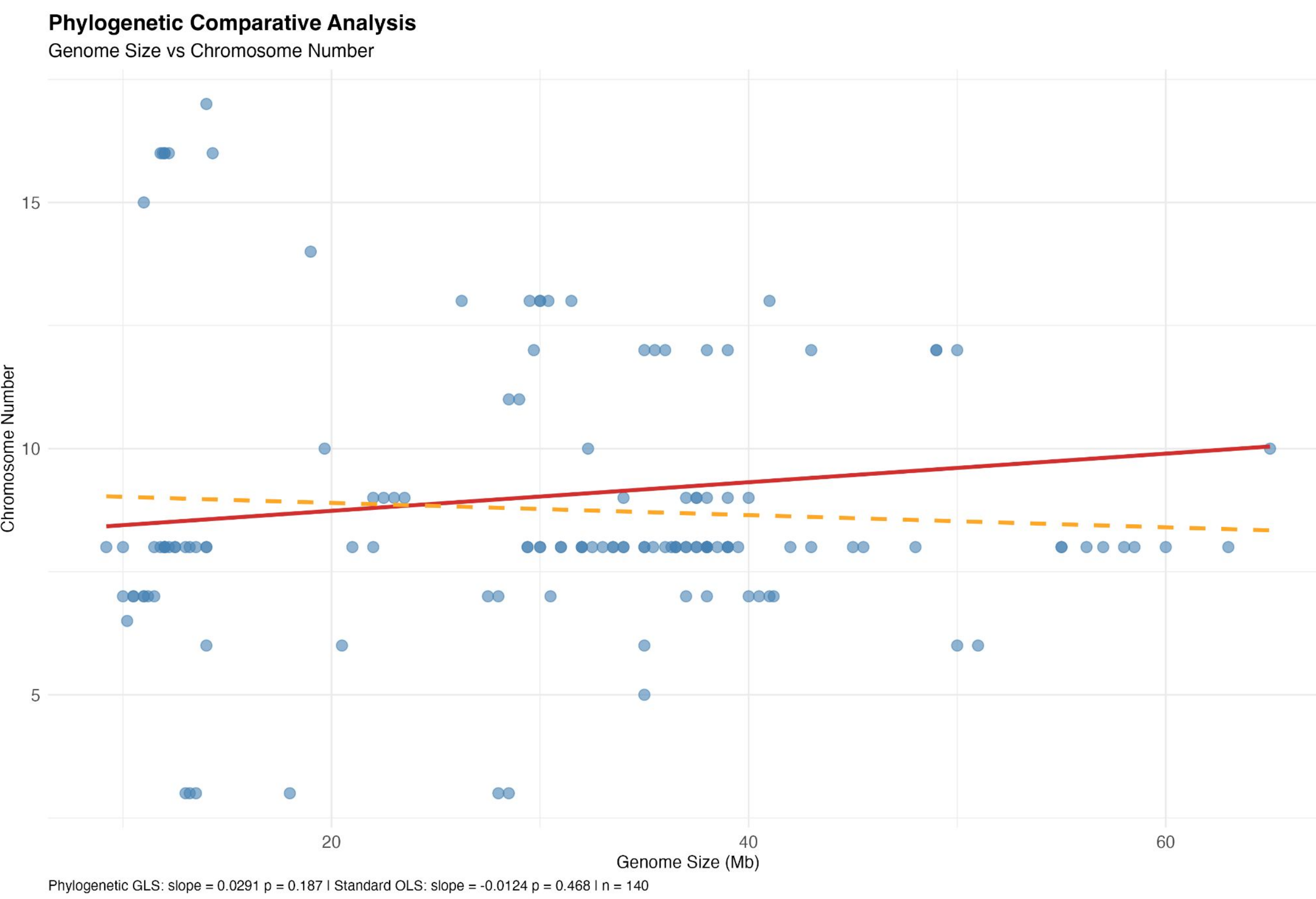


Figure 2: Generalized Least Squares (GLS) Regression Analysis

A scatter plot of genome size (x-axis, Mb) vs. chromosome number (y-axis) for 140 fungal species. The red line shows phylogenetic GLS regression ($\beta = 0.029$, $p = 0.187$), while the purple dashed line shows standard OLS regression ($\beta = -0.012$, $p = 0.468$) for comparison. Both relationships are non-significant. The slope reversal demonstrates why phylogenetic methods are essential analysis

Conclusion

This study found no significant relationship between genome size and chromosome number in fungi, indicating these genomic traits evolve independently. Both traits show strong phylogenetic signal, reflecting evolutionary constraints and lineage-specific solutions to genome organization. Rather than a simple linear relationship, fungal genome architecture appears to result from distinct evolutionary mechanisms operating on genome size and chromosome number separately. These findings highlight the importance of using phylogenetic comparative methods when analyzing species data and underscore the complexity of genomic evolution in fungi. Future research should investigate the specific mechanisms driving each trait independently and examine whether different fungal lineages follow different evolutionary rules.

Table 1: Statistical Analysis Summary

Analysis	Statistic	Value	p-value	Significant?
Pearson Correlation	r	-0.062	0.468	No
Spearman Correlation	ρ	0.052	0.540	No
OLS Regression	Slope	-0.0124	0.468	No
OLS Model Fit	R ²	0.0038	—	Explains 0.38% of variance
GLS Regression	Slope	0.0291	0.187	No
Phylogenetic Signal (Genome Size)	λ	0.9001	<0.001	Yes
Phylogenetic Signal (Chromosome #)	λ	0.9168	<0.001	Yes

